One Setpoint Step Certifies Robustness, with Application to Model-Free PID Tuning

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Abstract

The sensitivity peak M_s is the standard measure of robustness, but checking it on a running loop needs a model or an experiment that disturbs production. We show that one ordinary setpoint step suffices. For a stable digital loop with one degree of freedom, the increments of the error after a unit setpoint step are exactly the impulse response of the sensitivity function, so one finite record gives certified lower and upper bounds on M_s, without a model and under one assumption on the unrecorded tail. We apply this certificate to SPIN, a model-free PID tuner that raises the gains while the turns of the control error in three phase portraits stay below calibrated limits. The limits cannot certify robustness: a count measures setpoint crossings, and crossings bound M_s only on special loops. Accepting a tuning only if its certified upper bound is at most a target M^* makes SPIN provably robust, and the test never traps it. On 19 loops, published SPIN exceeded $M^* = 1.7$ on up to 9, certified SPIN on none.

1 Introduction

Let $L = CP$ be the loop transfer function and $S = 1/(1+L)$ the sensitivity function. Its peak $M_s = \max_\omega |S|$ measures robustness: near 1.4 is safe, near 2 is poor [2]. Tuning rules reach a chosen M_s with a model [2, 3]. Once a loop runs, its M_s is rarely checked [4, 5]: a check needs a model, a relay test, or a dedicated estimation from closed-loop data [6, 7, 8].

This paper shows that neither is needed. In a digital loop, the error after a setpoint step contains the sensitivity function exactly: its increments are the impulse response of S. One finite record of an ordinary setpoint step, or of any logged setpoint change made from rest, therefore certifies M_s without a model.

We use the certificate to make a model-free PID tuner provably robust. SPIN [1] tunes a PI or PID controller from one setpoint step per iteration, by counting how many turns the error makes in three phase portraits; its limits on the counts were calibrated, and its robustness was not guaranteed. The contributions are:

1. a certificate of M_s from one setpoint step (Section 2, Lemma 1, Theorem 1);
2. an explanation of why the turn counts cannot certify: a count measures setpoint crossings (Lemma 2), and crossings bound M_s only on special loops (Section 3.2, Proposition 1);
3. certified SPIN, which accepts a tuning only if its certificate meets a target M^*; it guarantees $M_s \le M^*$ whatever the limits and is never trapped (Section 3.3, Theorem 2).

Section 4 confirms the results in simulation. Unlike model-free methods that optimize performance from data [10, 11, 13, 12], the certificate bounds robustness and needs no extra experiment.

2 A Certificate from One Setpoint Step

A digital controller runs with the sampling period T_s. Seen at the samples, the loop it closes is linear and time invariant. Throughout, $L(z)$, $P(z)$, $S = 1/(1+L)$ and $M_s = \max_{\theta\in[0,\pi]} |S(e^{j\theta})|$, with $\theta = \omega T_s$, refer to this digital loop, the loop that actually runs.

Lemma 1 (The step contains S) *Let the loop be stable, have one degree of freedom, and have a strictly proper L (the measurement precedes the control action). Let e_n be the error after a unit setpoint step at $n = 0$, applied from rest and without disturbances, so that $e_{-1} = 0$ and $e_0 = 1$, and let $a_n = e_n - e_{n-1}$. Then*

$$S(e^{j\theta}) = \sum_{n\ge 0} a_n\, e^{-jn\theta}. \tag{1}$$

Proof: The z-transform of e is $S(z)/(1-z^{-1})$, so $S = (1-z^{-1})\mathscr{Z}\{e\}$. In a stable loop, a_n decays exponentially, so the series converges on the unit circle. □

A real record is finite, so one assumption covers what it does not show.

Assumption 1 (Long enough record) *The loop is stable, and the record $e_0, \dots, e_N$ satisfies $\sum_{n>N} |a_n| \le \tau$ for a known τ.*

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In a stable loop, $|a_n| \le c\rho^n$ with $\rho < 1$, so τ falls geometrically with the record length.

Theorem 1 (Certified bounds from one record) *Let the conditions of Lemma 1 and Assumption 1 hold, and let $\hat{S}(\theta) = \sum_{n=0}^{N} a_n e^{-jn\theta}$. Let G be a uniform grid on $[0, \bar{\theta}]$, $\bar{\theta} \le \pi$, that has spacing h and contains both ends. For any n_c, let*

$$J = \sum_{n=0}^{N} |n - n_c|\,|a_n|, \quad B = \sum_{n=1}^{N-1} |a_{n+1} - a_n| + |a_N| + 2\tau.$$

Then $\underline{M} \le M_s \le \overline{M}$, where

$$\underline{M} = \max_{\theta \in G} |\hat{S}(\theta)| - \tau,$$
$$\overline{M} = \max\left\{ \max_{\theta \in G} |\hat{S}(\theta)| + \tfrac{hJ}{2} + \tau,\ 1 + \tfrac{B}{\sin(\bar{\theta}/2)} \right\}, \qquad (2)$$

and the second term is omitted if $\bar{\theta} = \pi$. If τ equals the true remaining variation, $N \to \infty$, $h \to 0$ and $\bar{\theta} = \pi$, then both bounds converge to M_s.

Proof: By (1), $|S - \hat{S}| \le \sum_{n>N} |a_n| \le \tau$ at every θ, which gives $\underline{M}$. On $[0, \bar{\theta}]$, $|\hat{S}|$ equals $|e^{jn_c\theta}\hat{S}|$, whose derivative is bounded by J. Every θ lies within $h/2$ of a grid point, which gives the first term of $\overline{M}$. On $[\bar{\theta}, \pi]$, let $z = e^{-j\theta}$ and $Z_n = \sum_{m=1}^{n} z^m$, so that $|Z_n| \le 2/|1 - z| = 1/\sin(\theta/2)$. Summation by parts gives $\sum_{n\ge1} a_n z^n = \sum_{n\ge1} (a_n - a_{n+1}) Z_n$. Since $\sum_{n\ge N} |a_n - a_{n+1}| \le |a_N| + 2\tau$, we obtain $|S - a_0| \le B/\sin(\theta/2)$ with $a_0 = 1$. Convergence follows because, for fixed n_c, J stays bounded as $N \to \infty$, so $hJ \to 0$, and the remaining variation tends to zero. □

Everything in (2) except τ comes from the record, and all grid values follow from one chirp z-transform. The grid term covers the frequencies up to $\bar{\theta}$; above them, B bounds $|S|$ at once, and it is small for a smooth record. Fig. 1 shows the identity and the bounds on two loops.

3 Application to Model-Free PID Tuning

3.1 The SPIN method

SPIN [1] tunes a PI or PID controller with the gains K_i, K_p, K_d. In each iteration, it applies one setpoint step, records the error e, and forms three phase portraits,

$$\Gamma_0 : (e, \textstyle\sum e), \quad \Gamma_1 : (\Delta e, e), \quad \Gamma_2 : (\Delta^2 e, \Delta e), \qquad (3)$$

where $\sum e$ is the running sum of e and Δ the difference of consecutive samples. They correspond to the frequency bands of K_i, K_p and K_d, which we call bands 0, 1 and 2. Each axis is divided by its peak, and the count N_k is the number of turns that Γ_k makes around its end point before the error stays within $\pm\delta$ (2% of the step). Algorithm 1 states the rule.[1]

[1] An online simulator: `https://robopid-simulator.uceeb.cvut.cz`

3.2 Why the turn counts cannot certify

In each portrait (3), the second coordinate q is the running sum of the first coordinate p. The count is the net counterclockwise angle around the end point, placed at the origin, divided by 2π; we state the results in continuous time, where $\dot{q} = p$. For Γ_0, the zeros of $p = e$ are the setpoint crossings.

Lemma 2 (Half turns) *Let $\dot{q} = p$ with p continuous, let $p(t_1) = p(t_2) = 0$ with $q(t_1)q(t_2) < 0$, and let p not change sign on (t_1, t_2). Then, on $[t_1, t_2]$, the curve (p, q) turns counterclockwise around the origin by exactly half a turn, and so does its image under any linear map with a positive determinant, such as the per-axis normalization of SPIN.*

Proof: On (t_1, t_2), p keeps one sign, so the curve stays in one closed half-plane bounded by the q-axis, and q moves monotonically from one ray of the q-axis to the opposite ray; it avoids the origin, since $q = 0$ only where $p \ne 0$. The angle thus changes by exactly π, and counterclockwise, since q increases where $p > 0$ and decreases where $p < 0$. A linear map with a positive determinant maps half-planes to half-planes and opposite rays to opposite rays, and keeps the orientation. □

So each setpoint crossing adds half a turn to N_0, and how many crossings a count sees depends on how many rise above δ. Whether crossings reveal M_s depends on the loop. Lambda, IMC and SIMC tunings cancel the plant lags [3], e.g., PI with $T_i = T$ on $Ke^{-sL_d}/(Ts + 1)$; the loop is then

$$L(s) = \frac{k\, e^{-sL_d}}{sL_d}, \qquad k = \frac{KK_cL_d}{T_i}. \qquad (4)$$

Proposition 1 (Crossings on lag-cancelled loops) *The loop (4) is stable for $0 < k < \pi/2$, and its M_s strictly increases with k, as given by*

$$k = x\sin x - x^2\cos x, \qquad \frac{1}{M_s^2} = 1 - \frac{2k\sin x}{x} + \frac{k^2}{x^2}, \qquad (5)$$

with $x \in (0, \pi/2)$ the normalized peak frequency. The error after a setpoint step never crosses the setpoint if and only if $k \le 1/\mathrm{e}$, that is, $M_s \le 1.394$; otherwise it crosses infinitely often.

Proof: The phase of L is $-\pi/2 - x$ at $x = \omega L_d$; it reaches $-\pi$ first at $x = \pi/2$, where $|L| = 2k/\pi < 1$, and $|L| = k/x$ decreases, so the loop is stable. Let $h(x) = |1 + L|^2 = 1 - 2k\sin x/x + k^2/x^2$. Then $h'(x) = 2k(\varphi(x) - k)/x^3$ with $\varphi(x) = x\sin x - x^2\cos x$ increasing on $(0, \pi)$ and $\varphi(\pi/2) > k$, so h has one minimum $x^* \in (0, \pi/2)$ on $(0, \pi)$; on $[\pi, 2\pi]$, $h \ge 1$, and for $x \ge 2\pi$, $h \ge (1 - k/x)^2 > h(\pi/2)$. This gives (5), and by the envelope theorem $\frac{d}{dk}h(x^*) = -2\cos x^* < 0$. The error obeys $\dot{e}(t) = -(k/L_d)e(t - L_d)$ with $e = 1$ on $[0, L_d]$; it stays positive if and only if the characteristic equation $s + (k/L_d)e^{-sL_d} = 0$ has a real root, that is, if $k \le 1/\mathrm{e}$, and otherwise all solutions oscillate [9]. □

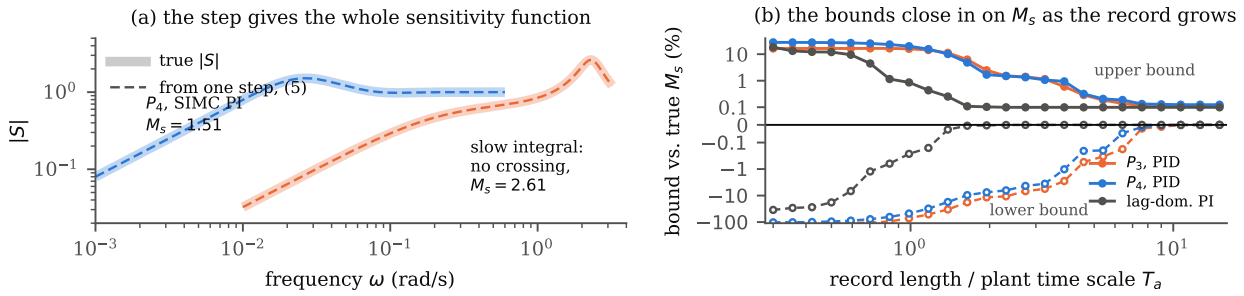


Figure 1: (a) $|S|$ from one setpoint step (dashed) lies on the true $|S|$ (thick) (Lemma 1), also for a slow-integral loop whose error never crosses the setpoint by more than 2%. (b) The bounds (2) relative to the true M_s versus the record length, with τ the true remaining variation (Theorem 1).

Algorithm 1: one iteration of SPIN [1] and, with lines 7a–7c in bold instead of line 7, of certified SPIN (Section 3.3)

1: apply a setpoint step and record the error e
2: **if** the record does not settle **then**
3: divide K_i, K_p, K_d by 2, 4, 8 (retreat)
4: **else**
5: count the turns N_0, N_1, N_2 of $\Gamma_0, \Gamma_1, \Gamma_2$
6: **if** $N_k \le \bar{N}_k$ for every band k **then**
7: raise all gains by one increment (accept)
7a: **compute $\overline{M}$ from the record (Theorem 1)**
7b: **if $\overline{M} \le M^*$ then** raise all gains (accept)
7c: **else lower the gains:** the band move that Prop. 2 predicts to lower M_s (highest such band, at most $n_{\max}$ in a row), else a uniform back-off (reject)
8: **else**
9: $k \leftarrow$ lowest band with $N_k > \bar{N}_k$
10: lower the gain of band k by one increment and raise the gains of the bands below k by one

An increment scales a gain by $1/(1-\beta)$. Defaults: $\bar{N} = (0.5, 0.75, 1.0)$, $\delta = 0.02$, $\beta = 0.1$; gains stay within $[0.001, 10]$ times their initial values (the gain box); the result is the last accepted tuning. Certified SPIN: in line 7a, $\bar{\theta}$ solves $1 + B/\sin(\bar{\theta}/2) = M^*$ ($\bar{\theta} = \pi$ if no solution) and $h = \bar{\theta}/\lceil \bar{\theta} J/(2\eta) \rceil$, so that $hJ/2 \le \eta$; $M^* = 1.7$, $\tau = 10^{-3}$, $\eta = 2 \cdot 10^{-3}$, $n_{\max} = 10$.

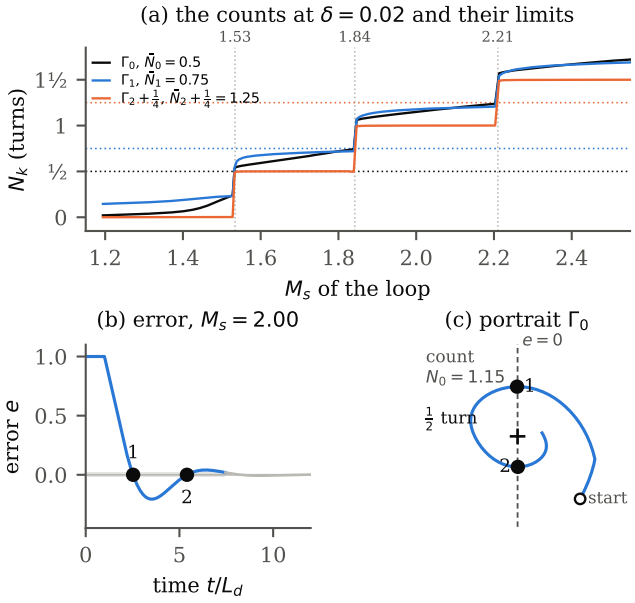


Figure 2: The lag-cancelled loop (4), SPIN's counts. (a) At $\delta = 0.02$, each count steps over its published limit (dotted) at $M_s = 1.53$, 1.84, 2.21; N_2 and its limit are drawn a quarter turn up (see text). (b) A record with $M_s = 2.00$ and two crossings. (c) Its portrait Γ_0 turns by exactly half a turn between them (Lemma 2); the radius is compressed, the angle is not.

At $\delta = 0.02$ the counts of the loop (4) therefore form a staircase in M_s (Fig. 2(a)). With n crossings above δ, the computed N_0 and N_1 lie within a quarter turn above $n/2$, and N_2 is $n/2 - 1/4$ within 0.01: on this ideal loop $\dot{e}$ jumps at $t = L_d$, so Γ_2 starts with a quarter turn clockwise. The published limits lie in the gaps: $\bar{N}_0 = 1/2$, $\bar{N}_1 = 3/4$ and $\bar{N}_2 = 1$ allow 0, 1 and 2 crossings above δ, that is, $M_s \le 1.534$, 1.842 and 2.210, where the $(n+1)$th extremum equals δ. Each band allows one crossing more than the band below, which is why the limits grow; Γ_0 binds, so on these loops SPIN stops at $M_s \le 1.534$.

On general loops, this guarantee is lost. The counts leave the staircase: in the 3800 count decisions of Section 4, N_0 and N_2 fell between $n/2 + 1/4$ and $(n+1)/2$ in 22 and 67% of them. And crossings do not bound M_s: a first-order plant with $T = 0.3L_d$ under PI with $T_i = 8T$ and a gain margin of 1.624 has $M_s = 2.62$, yet its error never crosses the setpoint by more than 2% (Fig. 1(a)), while a controller zero can cause crossings at a low M_s. The counts can shape a tuning, but only a certificate can certify it.

3.3 Certified SPIN

Certified SPIN accepts a tuning that the counts accept only if the upper bound $\overline{M}$ of Theorem 1, computed from the same record, is at most a target M^*. In Algorithm 1, it replaces line 7 by the bold lines 7a–7c. The counts shape the tuning; the

certificate decides whether it is accepted. A rejected tuning is lowered by a band move, as in line 10, or by a uniform back-off, the reverse of an acceptance.

Theorem 2 (Certified SPIN) *Let the plant be stable, let $K_iP(1) > 0$, and let every record tested in line 7a meet the conditions of Lemma 1 and Assumption 1.*

(i) Every tuning accepted in line 7b, and hence the result, has $M_s \le M^*$, whatever the limits $\bar{N}_k$.

(ii) For a controller C, let $\mu = \min\{0, \inf_{0<\theta\le\pi} \mathrm{Re}\, L(e^{j\theta})\}$. For $0 < \lambda < 1/|\mu|$, the loop with λC is stable, and

$$M_s(\lambda C) \le \frac{1}{1-\lambda|\mu|}. \tag{6}$$

(iii) If $M^* > 1+2\tau+\eta$ and no gain is clipped by the gain box, every run of rejections in line 7b ends in an acceptance or in a move of Algorithm 1 (line 3 or 10): it has at most $n_{\max}$ band moves, and each stretch of back-offs in it at most $\lceil \ln \frac{|\mu|M'}{M'-1} / \ln \frac{1}{1-\beta} \rceil$, where $M' = M^* - 2\tau - \eta$ and μ belongs to the controller at its start.

Proof: (i) This follows from Theorem 1. (ii) L is bounded on the unit circle except near $z = 1$, where the integral term has a bounded real part, because P is analytic at $z = 1$ and $\mathrm{Re}\, 1/(e^{j\theta}-1) = -\frac{1}{2}$; so μ is finite. Then $|1+\lambda L| \ge 1 + \lambda\, \mathrm{Re}\, L \ge 1 - \lambda|\mu| > 0$, which gives (6). The Nyquist curve of λL lies in $\mathrm{Re} \ge -\lambda|\mu| > -1$, also on a small indentation outside $z = 1$, where $K_iP(1) > 0$ keeps $\mathrm{Re}\, L > \mu$; it cannot encircle -1, and L has no poles outside the unit disc, so the loop is stable. (iii) Line 7c allows at most $n_{\max}$ band moves. After j back-offs from C, $\lambda = (1-\beta)^j$, and (6) gives $M_s \le M'$ for the stated j. In (2), the grid term is then at most $M_s + 2\tau + \eta \le M^*$, and the second term is at most M^* by the choice of $\bar{\theta}$; so the test passes unless the counts reject first. □

So the test never traps SPIN. A back-off is safe by (ii) but slows load rejection; a band move is often cheaper, and the same record predicts what it does.

Proposition 2 (Prediction from one step) *For any controller C' on the same plant, $S' = S/(S + (1-S)C'/C)$ exactly. If the record gives $\hat{S} = S + \varepsilon$, the predicted $\hat{S}'$ has the error $\hat{S}' - S' = \varepsilon\,(C'/C)/(D\hat{D})$, where $D = S + (1-S)C'/C$ and $\hat{D}$ is D with $\hat{S}$, wherever $D, \hat{D} \ne 0$. It predicts the frequency response of the new loop, not its stability.*

Proof: $L' = LC'/C$ with $L = (1-S)/S$; subtracting the two expressions gives the error. □

4 Simulations

Setup We compare published and certified SPIN. Both run 200 iterations with a derivative filter ratio of 10 and the gain box of [1]. Since the box ties the result to the start, every loop runs from SIMC on a half-rule model with τ_c three times its dead time (cautious start) and from the same tuning with all gains doubled (bold start). The 19 test loops are (1) the four plants of [1], $P_1 = e^{-s}/((10s+1)(s+1)^3)$, $P_2 = 1.25e^{-8s}/(5s+1)^4$, $P_3 = e^{-10s}/((2s+1)(s+1)^3)$ and $P_4 = e^{-4s}/(8s+1)^6$, with PI and with PID; (2) four plants that place the oscillation in a chosen band: the delay-dominant $e^{-5s}/(s+1)$ and the lag-dominant $e^{-s}/(10s+1)$ and $e^{-0.5s}/((10s+1)(s+1))$ with PI, and the fast $e^{-0.1s}/((s+1)(0.5s+1))$ with PID; (3) the family $e^{-L_d s}/((s+1)(0.5s+1)(0.25s+1))$ with PID and $L_d \in \{0.25, 0.5, 1, \ldots, 16\}$ (dead-time share 0.125 to 0.90). P, I and D act on the error. Let T_a be the dead time plus the lag time constants and T_f the derivative filter time constant; the controller runs at $T_s = \min(T_a/100, T_f/20)$ for PID and $T_a/1000$ for PI, with exactly discretized lags and the delay rounded to whole samples. Each record covers $15\,T_a$. All values of M_s are those of this digital loop, computed from its exact L. Because SPIN ends in a cycle between a few tunings, we report the most aggressive tuning accepted in the last 60 iterations.

Table 1: M_s of the digital loop for the most aggressive tuning that published and certified SPIN accept, from the cautious and the bold start, and the change of the load-step IAE (cautious start; –: tuning unchanged).

		published		certified		
		caut.	bold	caut.	bold	ΔIAE
P_1	PI	1.46	1.46	1.46	1.46	–
	PID	1.66	**1.84**	1.66	1.61	–
P_2	PI	1.64	1.67	1.64	1.67	–
	PID	1.59	**1.78**	1.59	1.66	–
P_3	PI	1.67	1.63	1.67	1.63	–
	PID	**1.81**	**1.74**	1.69	1.60	+23%
P_4	PI	1.55	1.60	1.55	1.60	–
	PID	1.61	1.55	1.61	1.55	–
delay-dom.	PI	1.68	1.63	1.68	1.63	–
lag-dom.	PI	**2.01**	1.69	1.67	1.69	+52%
2 lags	PI	1.44	1.47	1.44	1.47	–
fast	PID	**1.89**	**1.98**	1.66	1.69	0%

The certificate (Section 2) For all 146 tunings accepted below, a four times longer record showed a remaining variation $\le 2.8 \cdot 10^{-5}$, so Assumption 1 held with $\tau = 10^{-3}$; (1) held up to this variation, and every M_s lay within its bounds, with $\overline{M} - M_s \le \eta + \tau + 2.8 \cdot 10^{-5}$. Short records give wide bounds that still contain M_s (Fig. 1(b)).

The counts (Section 3.2) On the lag-cancelled loop, Γ_0 turned by half a turn between crossings within 10^{-6}, also after a linear map, and (5) matched a direct search for M_s to six digits.

Certified SPIN (Section 3.3; Table 1, Fig. 3)

- *Safety.* Published SPIN ended above $M^* = 1.7$ on 9 of the 19 loops from the cautious start (up to 2.01), on 8 from the

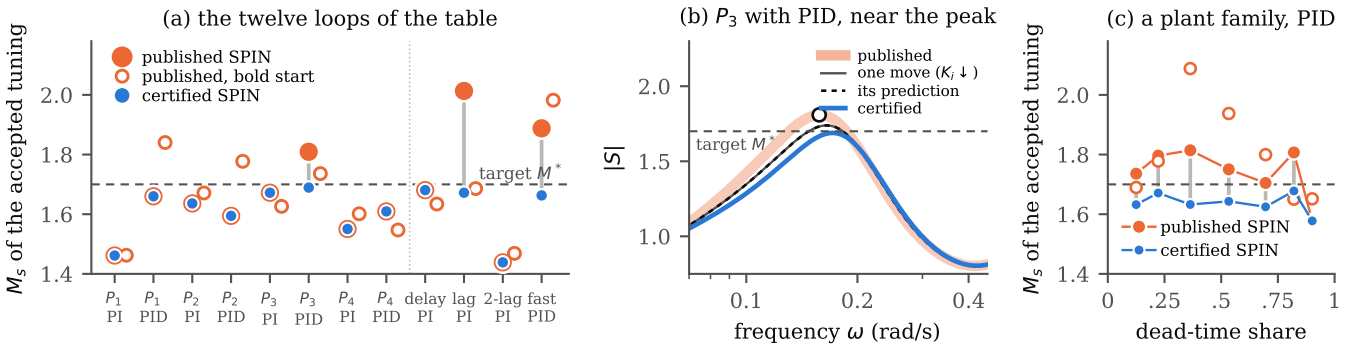


Figure 3: Published vs. certified SPIN, $M^* = 1.7$. (a) The loops of Table 1. (b) P_3 with PID: published tuning ($\overline{M} = 1.81$, ○), one band move lowering K_i ($M_s = 1.739$, as Proposition 2 predicts), certified tuning (1.689). (c) The plant family versus its dead-time share.

bold start (up to 2.09) and, without the gain box, on 12 of 18 (up to 2.05); certified SPIN on none (at most 1.69), as Theorem 2(i) guarantees.

- *No trapping.* (6) held along each back-off, predicted and true M_s after every band move agreed within 10^{-7}, and runs of rejections had at most 4 band moves; every loop kept accepting, except $L_d = 16$ without the box, where the counts lower K_d every round under either rule.
- *Cost.* The load-step IAE rose by 7.8% on average (6.4% from the bold start).

Limit values under the certificate With every limit $1/2$, certified SPIN still stayed below M^* (at most 1.68), but the counts did the rejecting (the certificate 14 and 8 times in the last 60 iterations, against 151 and 172): 2 and 3 loops accepted no tuning, and the load-step IAE rose by a median of 11 and 16%, up to sixteenfold (fast PID, $M_s = 1.21$). Raising $\bar{N}_1$ to 1 changed nothing, since N_1 never lay in $(3/4, 1]$. Under the certificate, the limits set performance, not robustness.

5 Conclusion

One ordinary setpoint step contains the sensitivity function of a digital loop, and one finite record certifies its peak without a model; so does every logged setpoint change made from rest. Applied to SPIN, the certificate turns a calibrated, model-free tuner into a provably robust one: the turn counts, which measure setpoint crossings and bound M_s only on special loops, keep shaping the tuning, and the certificate decides what is accepted. The theorems assume a linear loop and a long enough record; noise and real loops are left to a follow-up. Scripts: `https://github.com/ottapav/robopid-simulator`.

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